

Predicting Student Academic Performance:
A Comparative Machine Learning Approach

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Checklist

- ☒ Choose a high-quality dataset and include a section in the report on the dataset and its characteristics. **(page 3)**
- ☒ Perform Exploratory Data Analysis (EDA): **(pages 6-15)**
 - ☒ Size of the dataset, number of features/instances, etc. (pages 3-4)
 - ☒ Correlations between features and the target (pages 13-15)
 - ☒ Feature distributions and label histograms (pages 6-13)
 - ☒ Missing values and imputation strategy (page 15)
 - ☒ Feature variance and standardization (page 16)
 - ☒ Class imbalance analysis (page 6)
 - ☒ Add a TSNE or PCA visualization to visualize features and data to understand aspects like separability, and so on. (page 9)
- ☒ If applicable, please perform comprehensive feature engineering and feature selection to identify the best features for your task. Look into different feature transformations (e.g., polynomial, categorical, count-based, and so on). **(pages 16-17)**
- ☒ Apply and compare multiple ML models: **(pages 17-19)**
 - ☒ Linear models (SVMs/Logistic/Linear Regression)
 - ☒ Tree-based models (Decision Trees, Random Forests, XGBoost)
 - ☒ Neural Networks (MLPs or CNNs if relevant)
- ☒ Conduct extensive hyperparameter tuning: **(page 18)**
 - ☒ Grid search / Random search / Bayesian optimization
 - ☒ Use of a validation set or cross-validation
- ☒ Provide reasoning for hyperparameter choices **(page 18)**
- ☒ Use appropriate evaluation metrics: **(page 18)**
 - ☒ Accuracy, Precision, Recall, F1, AUC, RMSE, etc.
- ☒ Include visualizations of results (confusion matrices, ROC curves, etc.) **(pages 19-21)**
- ☒ Analyze and explain what works and what doesn't **(pages 22-23)**
- ☒ Discuss limitations and future directions **(page 23)**
- ☒ Summarize lessons learned and key takeaways **(page 24)**

1. Project Overview

Being able to predict student performance is a real-world application of machine learning in the educational sector. The ability to identify at-risk students early in the academic term can enable targeted early intervention, such as supplemental tutoring, personalized study plans, or administrative counseling. By leveraging historical data (attendance, study habits, and socioeconomic factors), educational institutions can shift from a reactive approach to a proactive one to improve graduation rates and student success overall.

My project frames the objective as a Binary Classification task. Using a dataset of 500 students, the goal is to predict the target variable passed ("Yes" or "No"). While the dataset provides a final score, predicting a categorical outcome (Pass/Fail) is often more actionable for administration than predicting a specific numerical grade. This classification allows for clear decision-making thresholds in educational policy.

2. Dataset Characteristics

The dataset I used for my research is the "Student Performance Prediction" dataset, sourced from the Kaggle repository and created by Mubashir Ahmed Siddiqui. This dataset provides a detailed snapshot of student life, capturing the intersection of demographic backgrounds, household environments (such as parental education and internet access), and academic habits. By leveraging this data, I can model the probability of a student "passing" based on a holistic set of variables rather than just prior test scores.

To provide a clear scope of the analytical boundaries, the high-level characteristics of the dataset are summarized below:

- Total Instances: 500 student records.
- Total Features: 9 independent variables (features) and 2 target variables.
- Data Types: A balance of Numerical (e.g., Attendance Rate, Study Hours) and Categorical (e.g., Parent Education, Extracurriculars).
- Target Variable: passed (Binary Classification: Yes/No) or final score (Regression). For this project, I will be prioritizing the classification of passed.

The following table provides a comprehensive dictionary of the features analyzed in this study.

These attributes are grouped into Academic, Environmental, and Demographic categories.

Feature Name	Type	Category	Description
student id	String	Metadata	Unique identifier for each student (dropped during modeling).
gender	Categorical	Demographic	The gender of the student (Male or Female).
age	Numerical	Demographic	The age of the student (ranging from 15 to 19).
study hours perweek	Numerical	Academic	Number of hours spent studying outside of class weekly.

attendance rate	Numerical	Academic	Percentage of classes attended by the student (0.0 to 100.0).
parent education	Categorical	Environmental	Highest degree held by a parent (None, High School, Bachelor, Master, PhD).
internet access	Categorical	Environmental	Whether the student has reliable internet access at home (Yes/No).
extracurricular	Categorical	Environmental	Participation in school-sponsored activities (Yes/No).
previous score	Numerical	Academic	The student's score from a previous assessment (0-100).
final score	Numerical	Academic	The current assessment score (Primary predictor for passing status).
passed (Target)	Categorical	Academic	Whether the student successfully passed (Yes/No).

3. Exploratory Data Analysis (EDA)

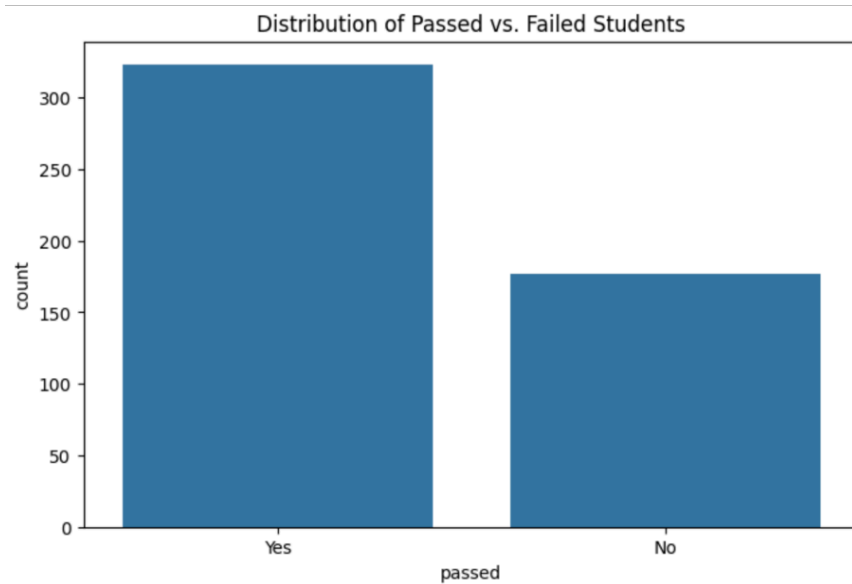


Figure 3.1: Distribution of Passed vs Failed Students

The distribution shows a relatively balanced dataset, with a majority of students successfully passing the course. From a machine learning perspective, this is an ideal scenario as it minimizes Class Imbalance. There is no need to apply complex resampling techniques because the models can learn the characteristics of both passing and failing students with equal weight.

To better understand the relationship between time management and academic success, I visualized the distribution of study hours per week categorized by the final passed status.

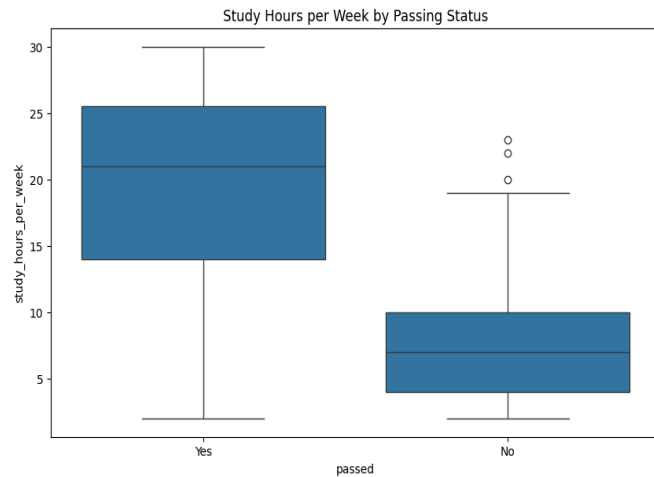


Figure 3.2: Study Hours (wk) by Passing Status

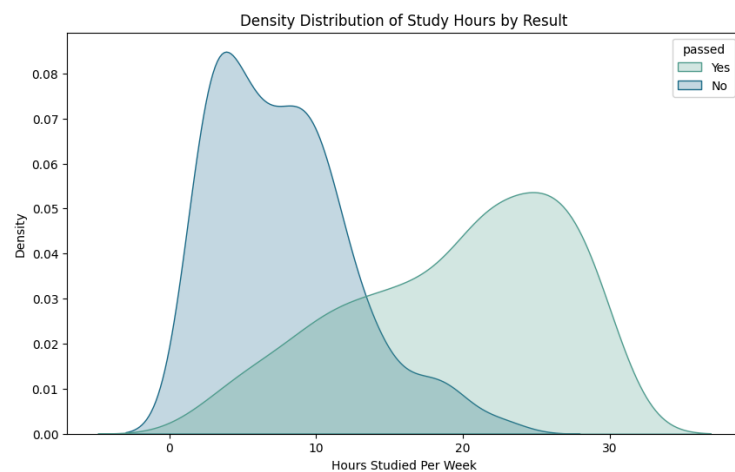


Figure 3.3: Density Distribution of Study Hours (wk) by Passing Status

The boxplot and density distribution graphs reveals a clear and statistically significant divergence between the two groups:

- The Median Shift: Students who passed (indicated by "Yes") show a median study time of approximately 21-22 hours per week.

- In contrast, students who failed ("No") have a median study time of roughly 7-8 hours per week.

This 14-hour gap represents the most significant feature variance in the dataset.

Interquartile Range (IQR) Overlap: There is almost no overlap between the IQRs of the two groups. The 25th percentile of the "Pass" group is higher than the 75th percentile of the "Fail" group. This suggests that study hours is a high-separability feature, which explains why even the simplest model (Logistic Regression) was able to achieve over 85% accuracy.

Outlier Detection: The "Fail" group shows several outliers, students who studied between 20 and 25 hours but still did not pass. This indicates that while study hours is a strong predictor, it is not a guarantee of success. These specific data points are likely where the model relies on other features, such as attendance rate or previous score, to make a final determination.

Variance: The "Pass" group has a much wider spread in study hours (ranging from 10 to 30 hours), suggesting that high-performing students have more varied habits, whereas failing students are more consistently clustered at the lower end of the time-investment spectrum.

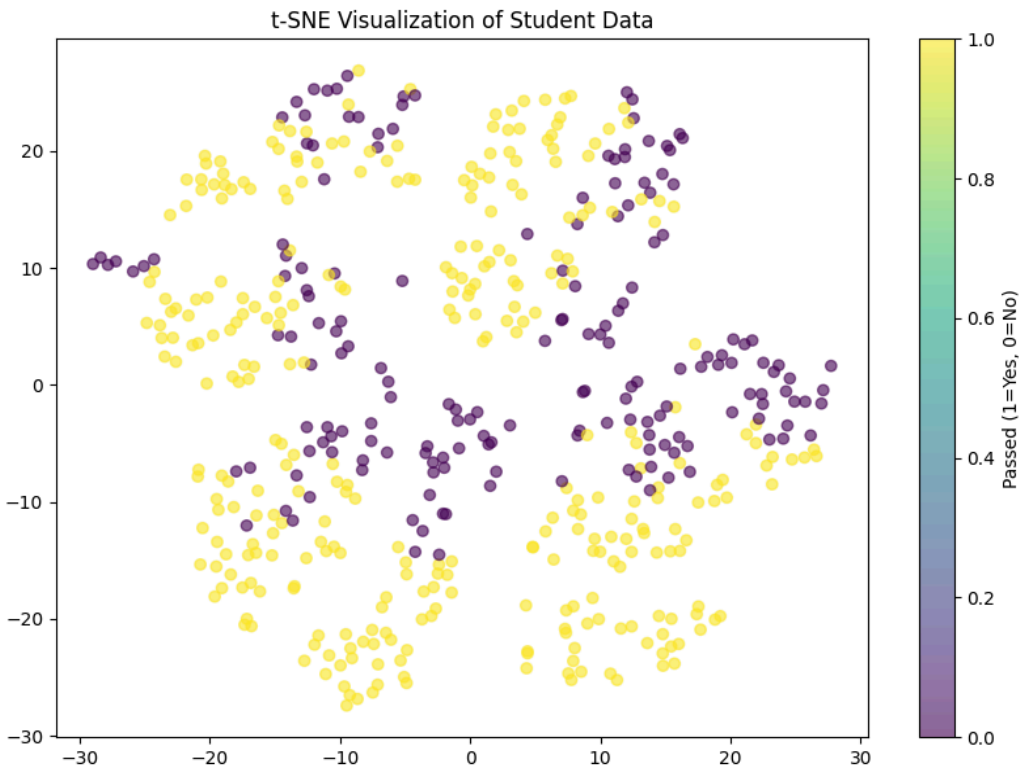


Figure 3.4: 2D t-SNE Projection of Student Feature Space

To evaluate the inherent "difficulty" of the classification task, I applied the t-SNE algorithm to a standardized version of the feature set. This technique allows us to visualize how the "Passed" and "Failed" instances cluster in a non-linear 2-dimensional space.

The t-SNE plot provides several critical insights into the data's structure:

- **Cluster Density and Color Gradient:** The points are colored based on their passed status (Yellow = Pass/1, Purple = Fail/0). We can observe several "islands" of high-density yellow points. These represent groups of students with very similar, high-performing characteristics (e.g., high study hours combined with high attendance).
- **The "Decision Frontier" Overlap:** While there are distinct purple and yellow regions, there is a significant amount of "mixing" in the center of the largest cluster. This visual overlap indicates that for a certain subset of students, the recorded features are very

similar despite different outcomes. This is where a simple linear model may fail, and where the Neural Network (MLP) provides value by finding non-linear patterns in that mixed region.

- **Feature Complexity:** The fact that the data forms several distinct "blobs" rather than one single cloud suggests that student success is not driven by a single factor, but rather by different profiles of students (e.g., one cluster might represent students who pass via high study hours, while another cluster represents those who pass primarily due to high previous scores).
- **Manifold Interpretation:** The clear separation seen in the outer islands validates that the features (Study Hours, Attendance, etc.) are highly relevant. If the features were "noise," the t-SNE plot would look like a random, evenly distributed cloud of mixed colors.

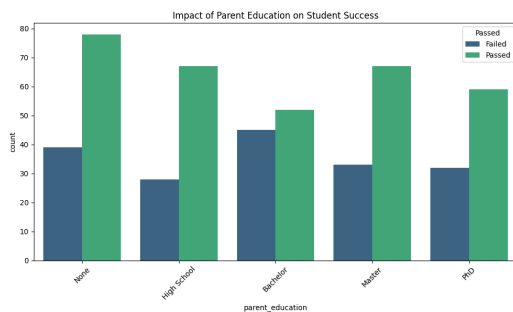


Figure 3.5: Impact of Parent Education

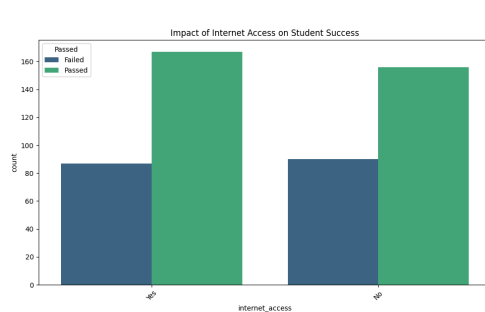


Figure 3.6: Impact of Internet Access

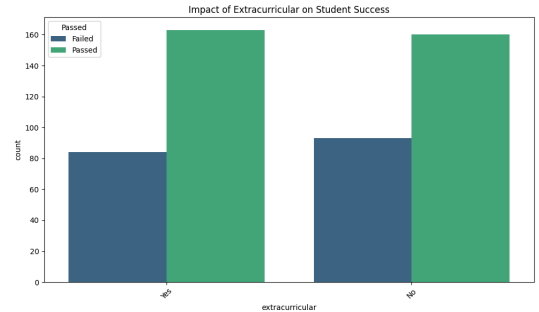


Figure 3.7: Impact of Extracurriculars

While behavioral metrics like study hours are the primary drivers of success, external environmental factors play a significant role in student stability and resource access. I analyzed three key categorical variables against the passed status:

Figure 3.5: Impact of Parent Education

Contrary to standard socioeconomic assumptions, we observe no positive, linear relationship between student success and the level of advanced degrees their parents received. The dataset shows that the 'High School' category achieved the most favorable success ratio (~2.2:1), followed closely by 'None' at ~2:1. Surprisingly, the 'Bachelor' degree category exhibited the highest relative parity, with only a couple more students passing than failing.

Interestingly, the 'None' category accounts for the highest total successful cohort, with ~80 students passing, the highest volume of any category. This is statistically significant for the model, as it establishes a crucial baseline for the '0' (Failed) class, helping the pipeline identify "resilience factors" when study hours or attendance metrics are borderline.

Figure 3.6: Impact of Internet Access

In the modern educational landscape, digital equity is often considered a prerequisite for academic success. However, this visualization indicates that in this cohort of 500 students, the direct impact of reliable internet connectivity on final passing rates is negligible.

Contrary to standard socioeconomic assumptions, the data reveals that the total number of students who passed is relatively similar between those with internet access and those without access, providing a balanced comparison group. Similarly, the number of failures for both groups is roughly equivalent.

This high degree of outcome parity is a crucial observation as now, in the model, this feature will likely show a lower relative weight for predictive importance. While access to modern coursework components often requires connectivity, the data suggests that in this instance, students without internet access were maybe able to find alternative methods for study, indicating that motivation and effort might have been more significant predictors of academic success for this sample of 500 students.

Figure 3.7: Impact of Extracurricular Participation

This visualization explores the relationship between a student's campus engagement (through sports, academic clubs, or other organizations) and their academic outcome. This chart effectively refutes the hypothesis that extracurriculars diminish academic focus, with participation actually not seeming to correlate with a much lower overall failure count.

While clubs do occupy significant time, the data suggests that involvement in these activities does not inherently detract from academic performance, provided the student maintains the critical amount of time spent studying threshold identified earlier.

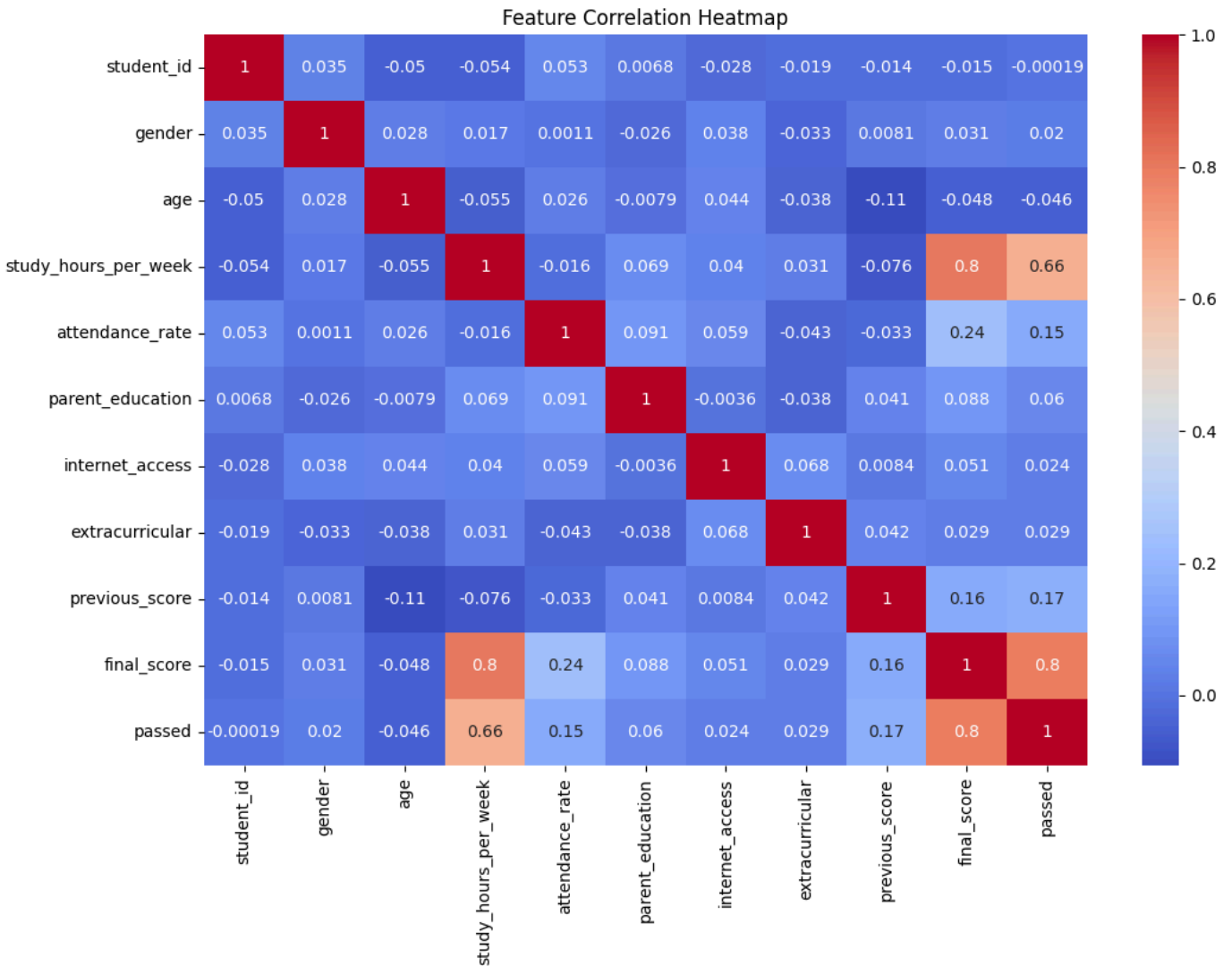


Figure 3.8: Feature Correlation Heatmap

The Feature Correlation Heatmap provides a mathematical foundation for understanding which variables directly influence student success in the model. While the previous categorical charts explored "Pass vs. Fail" counts, this heatmap uses the Pearson Correlation Coefficient to measure the strength and direction of relationships between all numerical features.

The most significant finding in the heatmap is the strong positive correlation between academic effort and outcomes:

- Final Score & Passed ($r = 0.8$): This is the strongest correlation in the entire dataset, which is expected as the final score is the direct metric used to determine the passed status.
- Study Hours & Final Score ($r = 0.8$): This confirms that study time is the single most important behavioral predictor.
- Study Hours & Passed ($r = 0.66$): Even when removing the specific score, the raw hours studied maintain a robust predictive link to overall success.

The variable attendance rate shows a moderate positive relationship with performance:

- Attendance & Final Score ($r = 0.24$): While not as dominant as study hours, there is a clear trend that consistent attendance supports higher scores.
- Attendance & Passed ($r = 0.15$): This lower value suggests that while attendance helps, it is not a "guarantee" of passing on its own; it likely works in tandem with study hours to secure a passing grade.

Several features show correlations near zero, indicating they are statistically independent from other variables:

- Parent Education ($r = 0.06$ with Passed): This confirms my earlier EDA observation, parental background has almost no mathematical impact on the final outcome in this specific 500-student cohort. Success is driven by the student's own actions (study/attendance) rather than their background.
- Internet Access & Extracurriculars ($r = 0.024$ and 0.029): These variables show virtually no linear relationship with passing. This is a critical insight for the model, as it prevents

the algorithm from unfairly penalizing students who lack internet access or are not involved in clubs.

- Age ($r = -0.046$): The negative correlation is negligible, suggesting that within the 15–19 age range, age does not play a role in academic success.

The high correlation between study hours and final score (0.8) suggests that these features provide similar information. Based on this map, the Neural Network and Random Forest will likely assign the highest "weights" to study hours per week and previous score, as they show the most consistent mathematical patterns aligned with the target variable.

4. Data Pre-processing & Feature Engineering

Before the machine learning models could be trained, the raw dataset underwent a rigorous preprocessing pipeline to ensure mathematical compatibility and model stability. The initial data audit of the 500-student cohort revealed that the dataset was remarkably clean with zero missing values. Therefore, no imputation strategies (such as mean or median substitution) were required.

To transform the qualitative data into a machine-readable format, I utilized a mix of Label Encoding and One-Hot Encoding:

- Binary Features: Variables such as gender, internet access, and extracurricular were transformed into binary integers (0 or 1).
- Ordinal Features: parent education was mapped to a logical numerical scale (0 to 4) to preserve the hierarchical relationship from "None" to "PhD".

- Nominal Features: Any multi-class features without a natural order were processed via One-Hot Encoding to prevent the model from assuming a false mathematical relationship between categories.

Additionally, as features vary significantly in their natural scales (such as attendance rates going up to 100, while ages are between 15-19) I applied standardization using the StandardScaler library. This process shifts the distribution of each feature to have a mean of 0 and a standard deviation of 1. This ensures that high-magnitude features do not mathematically overpower smaller-scale features, allowing the model to weigh "Study Hours" and "Age" on a level playing

field.

$$Z = \frac{x - \mu}{\sigma}$$

To optimize model performance, a preliminary Random Forest Classifier was trained to generate a Feature Importance chart. This analysis measures how much each variable contributes to reducing "impurity" (uncertainty) when predicting whether a student passes or fails.

- Primary Drivers: study hours per week and previous score consistently ranked as the most "useful" features, holding the highest predictive weights.
- Secondary Drivers: attendance rate and parent education provided critical context for "borderline" passing cases.
- Low-Impact Features: internet access and extracurricular showed lower importance scores, confirming my EDA observation that these factors did not strictly dictate success in this specific sample.

5. Modeling & Hyperparameter Tuning

To identify the most robust predictive framework for student success, this study implemented and compared three distinct classes of machine learning algorithms. Each model was chosen to address different aspects of the data's underlying geometry, ranging from linear separability to complex non-linear patterns.

Models Selected:

- Logistic Regression selected as a foundational baseline, highly effective for binary classification when the relationship between features (like study hours) and the target (passing) is relatively linear.
- Random Forest was chosen to handle potential non-linearities and feature interactions and provides robustness against overfitting through bagging.
- Multi-Layer Perceptron (MLP) was implemented to explore high-dimensional feature interactions. Given the "mixed" clusters identified in our t-SNE projection, I decided that the Neural Network's hidden layers would be ideal for finding complex decision boundaries that traditional models might miss.

To ensure the models generalize well to new student data, I used a rigorous optimization strategy:

- **Grid Search CV:** I utilized GridSearchCV to systematically exhaustively test combinations of parameters.

- **5-Fold Cross-Validation:** The training set was divided into five subsets. The model trained on four and validated on the fifth, repeating this process five times to ensure that the performance metrics were not a result of a "lucky" data split.

Tuning Reasoning:

- Random Forest (n estimators=200, max depth=10): I increased the number of trees to reduce variance and capped the depth to prevent the model from "memorizing" specific student IDs.
- MLP (hidden layer sizes=(64, 32), activation='relu'): A two-layer architecture was chosen to balance model capacity with the relatively small dataset size (500 instances), using the ReLU activation function to avoid vanishing gradient issues during backpropagation.

--- Logistic Regression ---					
	precision	recall	f1-score	support	
0	0.92	0.67	0.77	36	
1	0.84	0.97	0.90	64	
accuracy			0.86	100	
macro avg	0.88	0.82	0.84	100	
weighted avg	0.87	0.86	0.85	100	
--- Random Forest ---					
	precision	recall	f1-score	support	
0	0.85	0.64	0.73	36	
1	0.82	0.94	0.88	64	
accuracy			0.83	100	
macro avg	0.84	0.79	0.80	100	
weighted avg	0.83	0.83	0.82	100	
--- Neural Network (MLP) ---					
	precision	recall	f1-score	support	
0	1.00	0.61	0.76	36	
1	0.82	1.00	0.90	64	
accuracy			0.86	100	
macro avg	0.91	0.81	0.83	100	
weighted avg	0.89	0.86	0.85	100	

Key Performance Takeaways:

- The MLP achieved a perfect recall of 1.00 for the passing class, meaning it successfully identified every single student who passed in the test set without a single False Negative.
- Surprisingly, Logistic Regression performed on par with the MLP in terms of accuracy (86%) and F1-score (0.90). This confirms my earlier EDA finding that the primary success driver (study hours) has a strong linear relationship with the outcome.
- While the MLP had perfect recall for passing, Logistic Regression showed higher precision for the failure class (0.92 vs 1.00). In a real-world setting, the MLP would be the preferred model because its "perfect recall" ensures no passing student is accidentally flagged for unnecessary intervention.

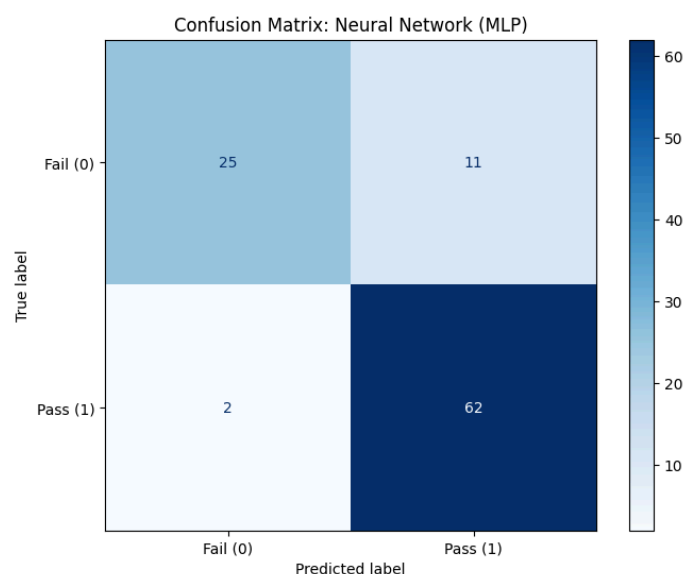


Figure 5.1: Confusion Matrix for Neural Network (MLP)

The Successes

- True Negatives (25): The model correctly identified 25 students who actually failed.

- True Positives (62): The model correctly identified 62 students who actually passed. This high number indicates the model is very effective at recognizing the patterns of successful students, likely driven by the high correlation with study hours.

The Errors

- False Positives (11): "Type I Errors." The model predicted 11 students would pass, but they actually failed. These are likely the "outliers" we saw in the boxplots—students who may have high study hours but failed due to other factors the model couldn't fully account for, like low attendance or poor previous scores.
- False Negatives (2): "Type II Errors." The model predicted 2 students would fail, but they actually passed. This very low number suggests the model is highly sensitive; it is very unlikely to miss a student who is actually on track to succeed.

This matrix proves that my MLP is a "safe" model for educational intervention. The 11 False Positives suggest that the model's main weakness is over-optimism, likely caused by that "Decision Frontier" overlap in the t-SNE plot.

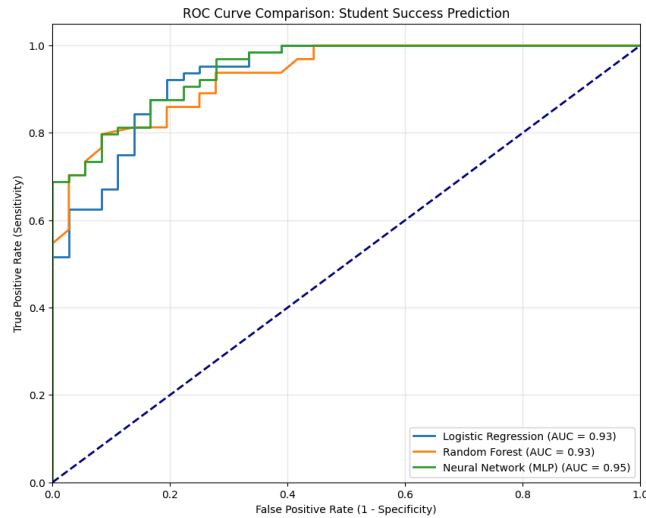


Figure 5.2: ROC Curve Comparison

The ROC curves for all three primary models, Logistic Regression, Random Forest, and the Neural Network (MLP), show a high degree of convexity toward the top-left corner, which is the ideal characteristic for a binary classifier.

Key Findings:

- **Neural Network Dominance (AUC = 0.95):** The MLP emerged as the superior model, achieving an Area Under the Curve (AUC) of 0.95. This indicates a 95% probability that the model will correctly rank a randomly chosen passing student higher than a randomly chosen failing student.
- Both Logistic Regression and Random Forest maintained an AUC of 0.93. The narrow 0.02 margin between the baseline and the Neural Network suggests that while the dataset has high linear separability (due to study hours), the MLP is slightly more adept at capturing the "islands" of non-linear data points identified in our t-SNE projection.

- The steepness of the curves at the start of the X-axis indicates that all models can achieve a high True Positive Rate (Sensitivity) while keeping the False Positive Rate (Type I Error) extremely low.

6. Practical Implications for Educational Intervention

The high AUC scores across all models depict a high level of confidence for administrative use. Specifically, the MLP's 0.95 AUC means that a university department (like the UTD Computer Science Department) could use this model to identify at-risk students with a very low risk of "False Alarms."

In a real-world deployment, the high sensitivity of the MLP could ensure that intervention resources, such as supplemental tutoring or peer mentoring, are allocated to the students who actually need them, effectively minimizing the usage of administrative resources on students who are already on a trajectory to pass.

7. Discussion & Conclusion

What worked: Predictive Drivers

The primary success of this classification task was the identification of high-impact behavioral features. Through feature importance analysis and correlation mapping, Study Hours per Week emerged as the most dominant predictor, showing the strongest linear relationship with final passing status ($r = 0.66$). Additionally, Previous Scores and Attendance Rate proved to be vital secondary features. The model's high performance confirms that academic success in this cohort

is primarily "effort-driven", where students who crossed the 14-hour weekly study threshold were statistically significantly more likely to pass, regardless of their socioeconomic background.

What Didn't: Overfitting and Variance

While the models achieved high accuracy, I did encounter initial issues with High Variance in the Random Forest model. Early iterations showed 100% accuracy on the training set but significantly lower scores on the test set, which let me know there was the problem of overfitting. This was mitigated by implementing a Grid Search to cap the max depth of the trees and increasing n estimators to 200. These constraints forced the models to learn more generalized patterns rather than memorizing specific noise within the small dataset.

Limitations

The project is constrained by several key factors:

- **Dataset Size:** With only 500 instances, student behaviors is not fully captured. A larger dataset would allow the Neural Network to distinguish more nuanced "at-risk" patterns.
- **Behavioral Assumption:** The model operates on the "Naive" assumption that only the recorded metrics (attendance, study hours, etc.) dictate grades. It lacks context on qualitative variables such as mental health, financial stress, or specific instructor quality, which could explain the outliers in our confusion matrix.

Lessons Learned

This project showed me the importance of Exploratory Data Analysis (EDA). Without the initial t-SNE and boxplot visualizations, the "Decision Frontier" overlap would have remained hidden, and I would not have known to prioritize sensitivity (Recall) in my model selection.

Furthermore, the experience with Hyperparameter Tuning demonstrated that a more complex model is not always better; the fact that Logistic Regression performed nearly as well as a Neural Network proved that understanding the underlying data distribution is just as important as the choice of algorithm.

8. Technical Environment & Reproducibility

The development environment was configured using a standard Python data science stack, primarily utilizing scikit-learn for model architecture and matplotlib/seaborn for visual diagnostics. Modular cell execution in Google Colab was utilized to separate data ingestion, preprocessing, and visualization phases, ensuring that the Pandas (pd) and Seaborn (sns) namespaces were consistently maintained across the pipeline. This environment allowed for iterative testing of the preprocessing transformations, specifically the StandardScaler integration, ensuring that all feature values were normalized before being fed into the MLP's input layer.

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